

UC LINA 600 LINEAR ALGEBRA FOR GRAPHICS · MODULE 09 · LECTURE NOTES

Determinants and Degeneracy

One number that tells you volume, orientation, and invertibility.

MODULE OUTCOMES

- Explain the determinant as a volume scale factor.
- Explain what a zero determinant means.
- Explain what a negative determinant means.
- Relate determinants to invertibility and solving.
- Use determinants in graphics contexts.

1 What it measures

THE DETERMINANT IS THE FACTOR BY WHICH THE TRANSFORM SCALES VOLUME — AND ITS SIGN SAYS WHETHER ORIENTATION FLIPPED

Apply the transform to a unit cube; the determinant is the signed volume of the parallelepiped that comes out — which is the whole geometric content, and everything algebraic about determinants follows from it.

So a determinant of 2 doubles volume and a determinant of 1 preserves it — and every rotation has determinant exactly 1, which is one way of saying that rotations do not distort (§3).

A negative determinant means the orientation was flipped — a reflection is involved — and the cube came out inside-out, which is §3's subject and has direct consequences for normals.

And zero means the cube was flattened entirely — collapsed into a plane, a line, or a point — which is §2, and is the case that matters most in practice because it is the one you have to test for.

2 Zero determinant

- **The volume collapsed to nothing** — the transform squashed three-dimensional space into something lower-dimensional, and the output has no thickness.
- **The columns are linearly dependent** (Module 02 §2) — **they do not span the space**, so the image of the transform is a proper subspace and most vectors are unreachable.
- **The matrix is not invertible** — **information was destroyed and cannot be recovered**, because many different inputs map to the same output and nothing can tell them apart afterwards.

- **The system $Ax = b$ has either no solution or infinitely many — never exactly one —** which is the linear-systems consequence and the one a solver reports.
- **And in graphics: a degenerate triangle, a collapsed bone in a skeleton, a singular model transform — all of which produce NaNs downstream** (Module 06 §2's zero-length normal). **Zero determinant means information was destroyed, which is why it cannot be inverted, and why all five statements are really the same statement.**

3 Sign and handedness

A NEGATIVE DETERMINANT FLIPS HANDEDNESS, WHICH IS HOW A MIRRORED OBJECT TURNS ITS NORMALS INSIDE OUT

Mirror an object by scaling one axis by -1 , and the determinant of that transform is -1 — so the effective winding order of every triangle reverses and every face now points inward (Module 06 §2).

Which is why a mirrored mesh renders inside-out unless the renderer notices and flips its backface culling to match — **and is a real and common asset-pipeline bug**, usually appearing when somebody mirrors a character to make its other half.

And it is why a rotation has determinant exactly $+1$: rotations preserve both volume and handedness, which is what distinguishes them from general orthogonal matrices (an orthogonal matrix with determinant -1 is a reflection or a rotation-plus-reflection).

So the test for 'is this actually a rotation' is: orthogonal, and determinant $+1$ — two conditions, both cheap to check — and Module 13 §2 uses exactly these two as a verification on any matrix claiming to be a rotation.

4 Using it

Use	What the determinant does there
Triangle area and orientation	The 2×2 determinant of the two edge vectors gives twice the signed area.
Backface culling after projection	The sign of that signed area says which way the triangle faces in screen space.
Barycentric coordinates	Ratios of sub-triangle determinants — which is why they are cheap to compute during rasterisation.
Ray-triangle intersection	Cramer's rule on a 3×3 system, which is the Möller-Trumbore algorithm's core.
Degenerate transform detection	A near-zero determinant before attempting an inverse — with the caveat in the closing callout.
Change of volume under a map	The Jacobian determinant, which is how a change of variables rescales a probability density (CSCE 647).

Never test `determinant == 0` in floating point — test against a tolerance, and prefer a condition number when the decision actually matters (the closing callout). The Jacobian row is what makes change of variables work in Monte Carlo integration: when you sample in one parameterisation and need a density in another, the Jacobian determinant is the correction factor, and getting it wrong biases the whole estimate (CSCE 647 Module 07).

AND THE COMPUTATIONAL WARNING

Cofactor expansion is $O(n!)$ and is unusable beyond about 4×4 — which is perfectly fine for graphics, where everything is 4×4 , and is a trap in every other setting, including anything in CSCE 633 or CSCE 669.

Real implementations use LU decomposition and multiply the diagonal, in $O(n^3)$ — and that is what a numerical library is doing when you ask it for a determinant.

And a tiny determinant does not mean nearly singular: scaling a 3×3 matrix by 0.01 scales its determinant by a millionth — without making it any harder to invert at all, since the inverse just scales back.

So for the question 'is this safe to invert', use the condition number rather than the determinant — which is the right tool for that question and is what numerical libraries report when they warn you about a matrix.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
3Blue1Brown, chapters 6 and 7 (free)	§§1 to 3 — the determinant as a volume factor, which is the framing this module uses throughout.
MIT 18.06, the determinant lectures (free video)	§§1 and 2 — the properties, and the connection to invertibility.
Möller & Trumbore — Fast ray-triangle intersection (free)	§4's fourth row — determinants doing real work in a renderer's inner loop.
Trefethen & Bau — Numerical Linear Algebra	The closing callout — condition numbers and why determinants are the wrong tool for conditioning. Library copy.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Compute the determinant** of five 2×2 and three 3×3 matrices by hand.
- 2 **Verify the volume interpretation** by transforming a unit cube.
- 3 **Construct a transform with determinant 2** and confirm the volume.
- 4 **Construct one with determinant 0** and show the columns are dependent.

- 5 **Try to invert it** and observe what your library does.
- 6 **Mirror a mesh** and observe the inside-out rendering.
- 7 **Verify a rotation matrix** is orthogonal with determinant +1.
- 8 **Compute a triangle's signed area** and use it for backface culling.
- 9 **Scale a matrix by 0.01** and compare determinant to condition number.
- 10 **Implement Cramer's rule** for ray-triangle intersection.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What does the determinant measure, geometrically?
- 2 What does its sign mean?
- 3 Give five equivalent consequences of a zero determinant.
- 4 Why are they all the same statement?
- 5 Why does a mirrored mesh render inside-out?
- 6 What two conditions test for a rotation?
- 7 Name four graphics uses of determinants.
- 8 Why should you never test for exact zero?
- 9 Why is cofactor expansion unusable at scale?
- 10 Why is a tiny determinant not the same as nearly singular?